

MetaBio - The Biomedical Intelligence Engine

Target Industries: Biotech, Pharmaceuticals, Drug Discovery, Life Sciences Research Institutes

<https://www.youtube.com/watch?v=ZHIIOWdziKQ>

1. Executive Summary

The biomedical sector is generating unprecedented volumes of data, but much of it remains underutilized. MetaBio is an advanced biomedical intelligence engine that transforms fragmented data into actionable knowledge. Designed for pharmaceutical and life sciences organizations, it enables teams to accelerate drug discovery, uncover adverse drug events (ADEs), and generate new scientific hypotheses with speed and precision.

At its core, MetaBio fuses natural language processing, biomedical ontologies, and graph-based data modeling to create a unified knowledge environment. This allows researchers to connect the dots across scientific literature, clinical trial data, and proprietary research assets—surfacing critical relationships between genes, diseases, drugs, and phenotypes.

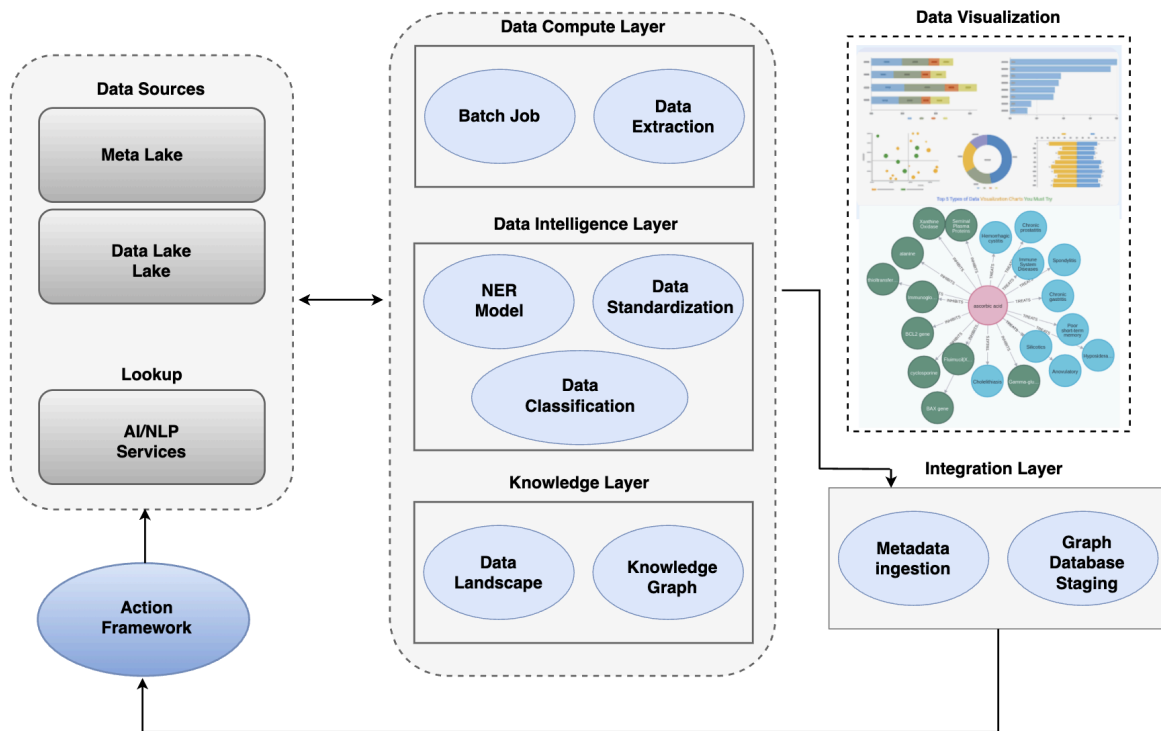
MetaBio is already powering high-impact use cases across the drug development lifecycle: identifying novel therapeutic targets, predicting off-target effects, and flagging potential ADEs early—long before they manifest in clinical trials. Its ability to semantically normalize and link diverse data sources makes it an indispensable engine for translational research, pharmacovigilance, and real-time scientific discovery.

2. The Business Imperative

Pharma and biotech organizations face a critical bottleneck: the inability to unify and make sense of the vast, scattered biomedical data generated internally and published externally.

- **Siloed & Inconsistent Data Landscapes:** Biomedical data lives in disparate formats—journal articles, clinical trials, internal reports, structured databases—each with inconsistent terminologies and taxonomies. This fragmentation inhibits data reuse and cross-functional collaboration.
- **Manual Curation Bottlenecks:** Research teams often rely on labor-intensive, manual processes to extract and standardize information. This slows discovery, drives up costs, and limits scalability.
- **Missed Insights & R&D Inefficiencies:** When critical data points are buried in unstructured text or disconnected systems, valuable insights go unnoticed.

This leads to longer development cycles, missed opportunities for drug repositioning, and suboptimal decision-making.



3. The MetaBio Solution: From Fragmentation to Federated Intelligence

MetaBio transforms biomedical complexity into clarity through a modular, intelligent, and highly secure Azure-based architecture. Built to unify scattered datasets across research publications, internal reports, clinical trial documents, and proprietary scientific data, the platform turns raw information into a dynamic and queryable biomedical knowledge graph.

At the heart of MetaBio lies an orchestrated processing pipeline powered by **Azure Logic Apps**, which acts as the central control layer for ingesting and managing biomedical data. Incoming queries or documents—whether structured or unstructured—are routed through **Azure API Management**, authenticated securely via **Azure Active Directory B2C**, and then passed to downstream services for semantic analysis and entity extraction.

All data—whether research corpora, raw biomedical files, or clinical datasets—is securely stored in **Azure Blob Storage**, serving as both a retrieval source and long-term repository. From here, **Azure Logic Apps** invokes specialized microservices for **machine learning model execution** (via Azure Machine Learning or custom endpoints) and **deep biomedical NLP inference**.

MetaBio then uses **Azure Cognitive Search** to intelligently index the processed information, enriching it with domain-specific ontologies and semantic tags. This enables high-precision, context-aware search experiences that allow scientists and researchers to surface connections between diseases, genes, compounds, and phenotypes—without needing to know exactly where or how that knowledge is stored.

The frontend application, served globally via **Azure Front Door and Static Web Hosting**, allows research users to intuitively interact with the knowledge layer in real time. Responses, insights, and results are piped back securely through the same orchestrated flow—ensuring performance, scalability, and traceability.

This Azure-native, federated architecture not only reduces reliance on manual curation but creates a continuous intelligence loop that accelerates scientific discovery across domains like drug development, adverse drug event identification, target discovery, and precision therapeutics.

Key Features & Capabilities:

- **AI-Powered Information Retrieval:** MetaBio scans and indexes large corpora of biomedical literature, clinical trials, patents, and proprietary data repositories. It ensures high recall of relevant information, regardless of source or format.
- **Deep NLP for Entity Extraction:** Our engine identifies biomedical entities—e.g., genes, pathways, compounds, phenotypes—with high precision, using transformer-based models trained on domain-specific corpora.
- **Ontology-Based Standardization:** Extracted terms are normalized using established biomedical ontologies such as UMLS, SNOMED CT, and MeSH. This harmonization is critical for enabling coherent queries across heterogeneous datasets.
- **Entity Linking & Disambiguation:** MetaBio resolves synonyms and contextual ambiguities by aligning entities to reference knowledge bases. This semantic enrichment enables reasoning over biomedical relationships.
- **Knowledge Graph Construction:** All entities and relationships are stored in a scalable GraphDB, enabling dynamic exploration of biological mechanisms, drug targets, adverse events, and more.
- **Exploratory Reasoning & Querying:** The end-user experience centers on intuitive, context-aware queries across the graph, revealing unexpected connections and supporting novel hypothesis formation.

4. Regeneron Use Case: Transforming Search into Discovery

Regeneron Pharmaceuticals adopted MetaBio to unify its biomedical data and streamline research workflows across departments. Facing challenges with disparate data sources, inconsistent terminologies, and fragmented internal knowledge, Regeneron leveraged MetaBio to:

- Ingest and harmonize scientific literature, internal research, and clinical trial data
- Extract and normalize critical biomedical entities using MetaBio's NLP pipeline
- Align data with industry-standard ontologies for cross-team consistency
- Connect datasets into a coherent graph structure to surface hidden relationships
- Enable researchers to query, visualize, and explore biomedical knowledge in real time

Outcome: A centralized biomedical intelligence layer that reduced manual curation, accelerated insight discovery, and empowered scientists with a unified source of truth for hypothesis generation and translational research.

5. Core Technology Architecture

MetaBio's architecture is purpose-built for the demands of modern biomedical R&D—scalable, interoperable, and semantically aware.

- **Information Ingestion Pipeline:** Automated ETL workflows collect and preprocess data from public and proprietary sources, including PubMed, ClinicalTrials.gov, patents, and internal reports.
- **Natural Language Understanding (NLU):** A domain-specific NLP stack powered by transformer models fine-tuned on biomedical corpora, enabling high-fidelity entity extraction and contextual comprehension.
- **Ontology & Taxonomy Integration:** MetaBio integrates with biomedical standards (UMLS, SNOMED CT, MeSH, GO, etc.) to drive consistency, interoperability, and semantic depth.
- **Graph Database Layer:** A high-performance GraphDB stores entities as nodes and relationships as edges. This facilitates fast, expressive queries that mimic how researchers think—via connections.
- **Security & Access Controls:** Role-based access, encryption at rest and in transit, and audit trails ensure data integrity and confidentiality—critical for pharma-grade deployments.

6. The Future of Biomedical Discovery

MetaBio is more than a tool—it's a strategic asset for the next generation of life sciences innovation. By converting unstructured data into structured knowledge, it

enables faster, smarter, and more connected research. With successful deployments like Regeneron's, MetaBio is rapidly becoming the standard for organizations seeking to operationalize biomedical intelligence at scale.

In a world where knowledge drives competitive advantage, MetaBio turns information overload into a catalyst for discovery.